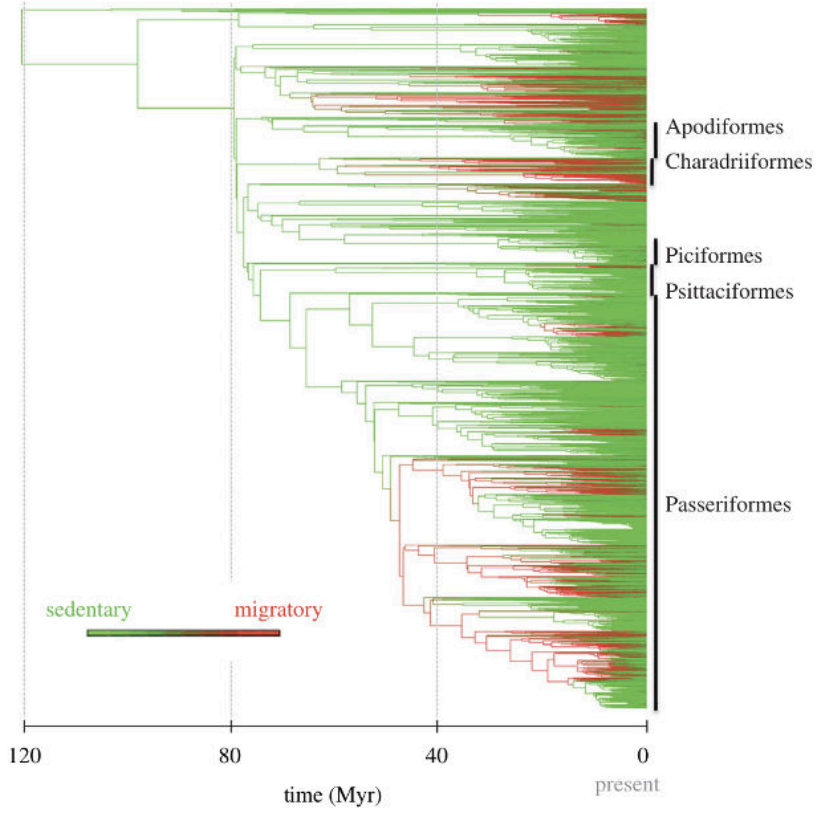


Migration

Dr. Ethan Linck
Office: 305 Lewis Hall

Migration

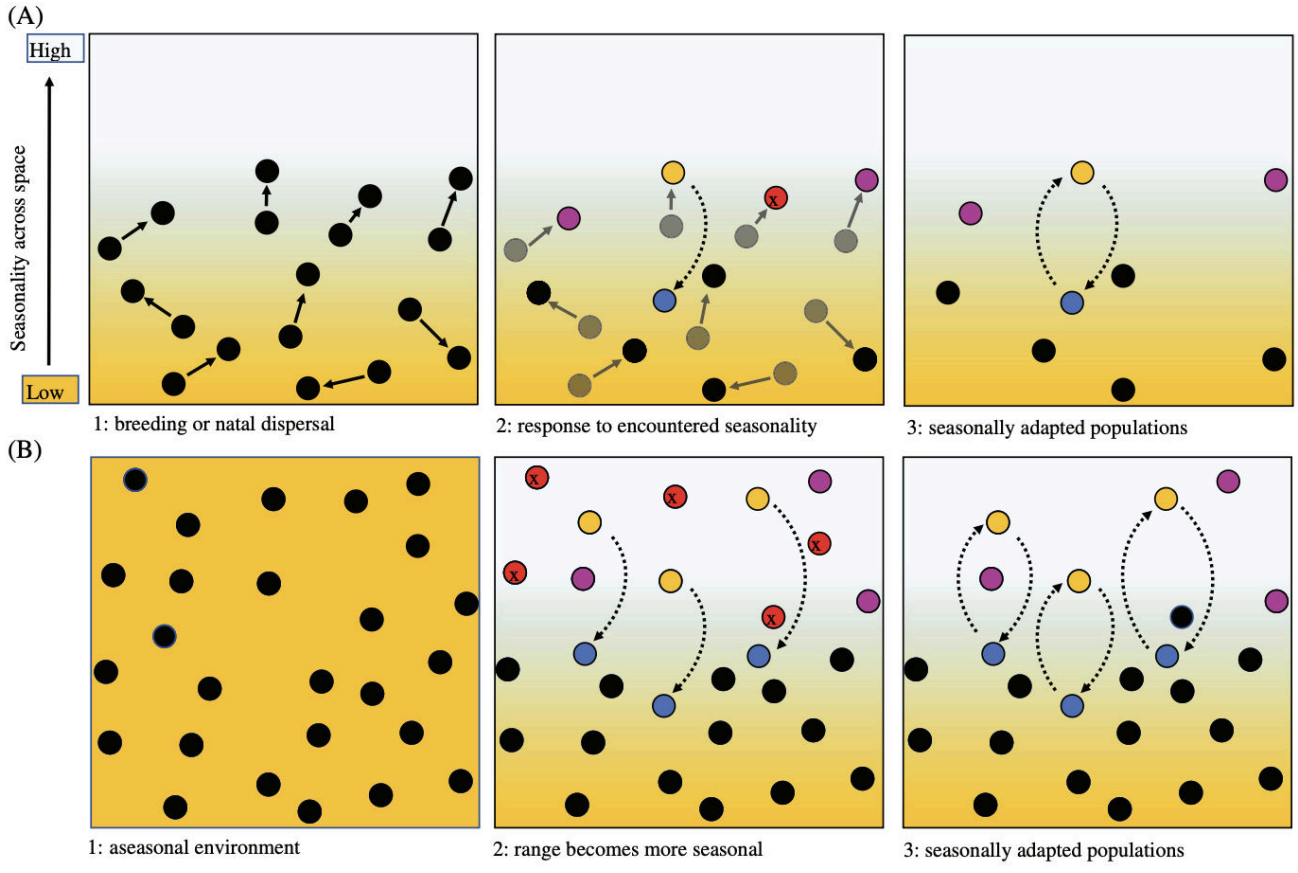
- 1817 bird species are migratory (~18%)
- Repeated evolution of migratory behavior: predictable seasonal movement across latitude, longitude, elevation
- c.f. **irruptive** movements (unpredictable search for resources)



(Rolland et al. 2014)

Migration

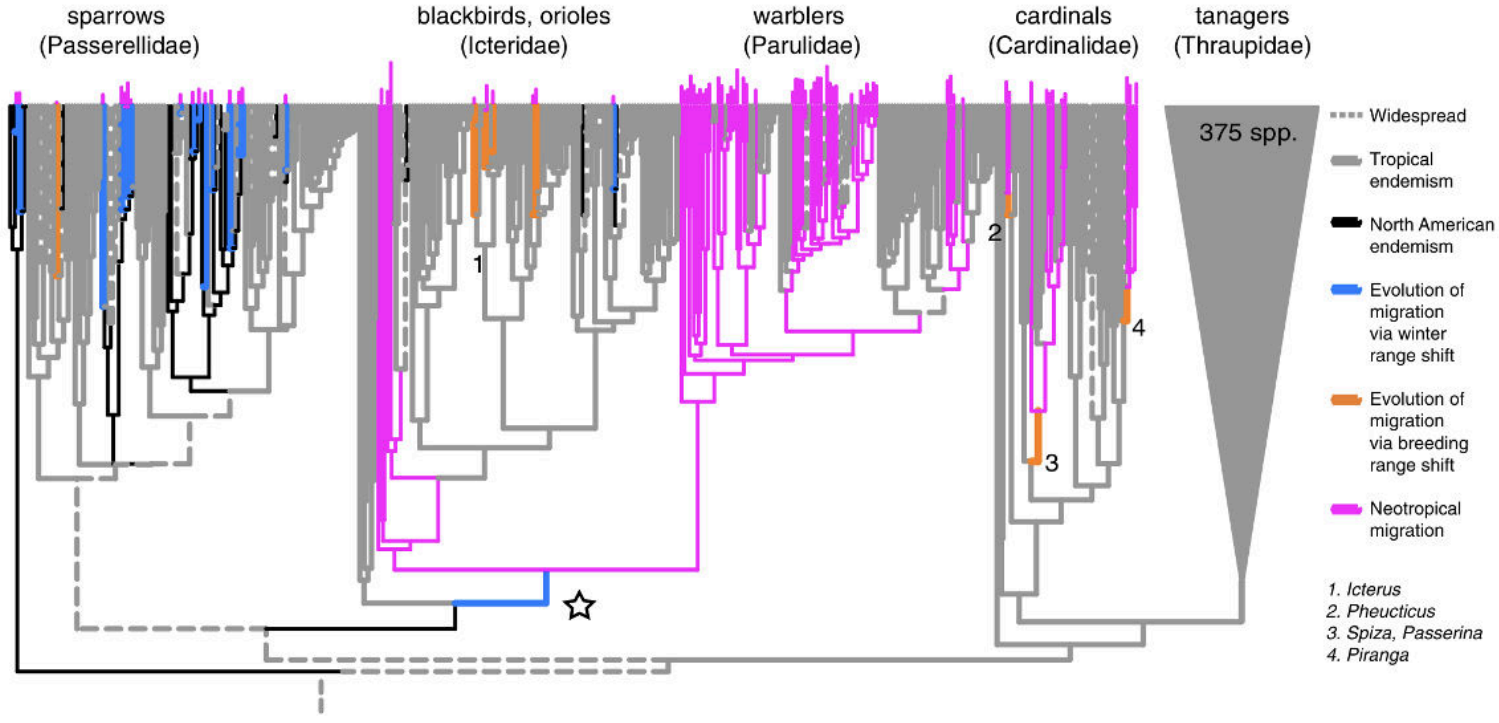
- H_1: Migration is adaptation to reduced competition for breeding sites
- H_2: Migration is adaptation to seasonality



(Rolland et al. 2014)

Which hypothesis is supported in neotropical migrants?

blue means a lineage with a winter range shift; orange means a lineage with a breeding range shit; pink are lineages that are current neotropical migrants

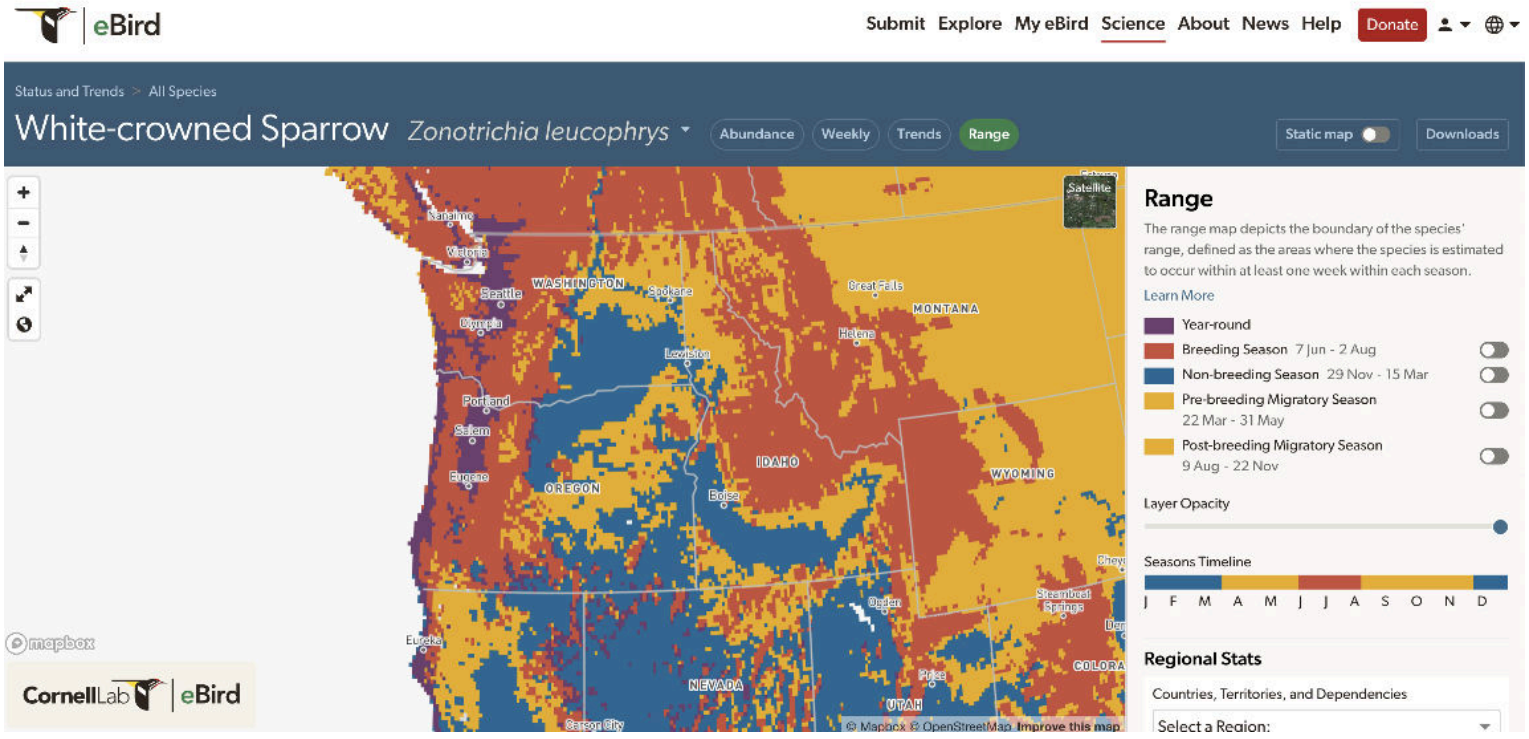


(Winger et al. 2014)

a) H_1: adaptation to reduced competition for breeding sites
b) H_2: adaptation to seasonality

Partial Migration

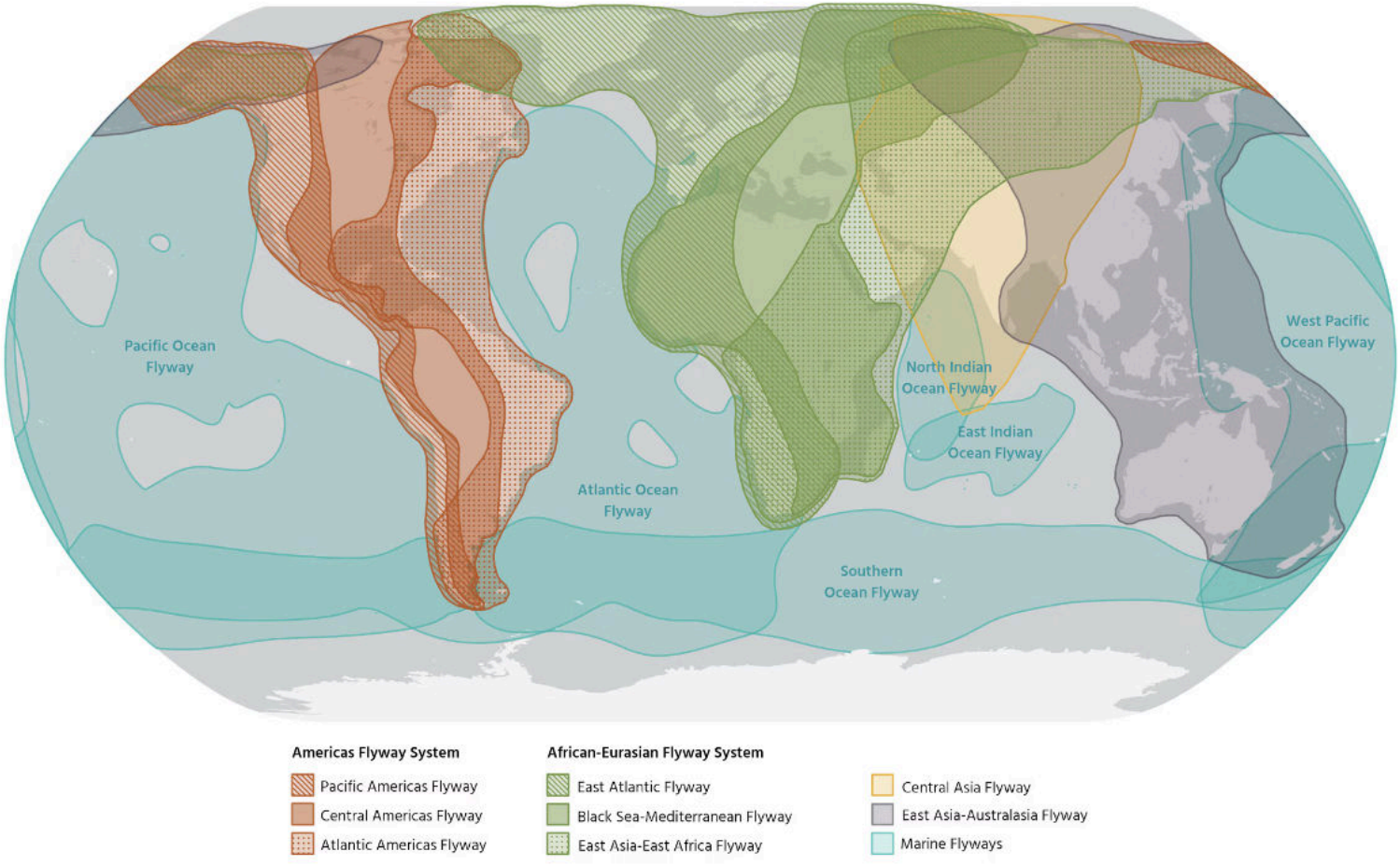
- Common: Found in (e.g.) 32-44% of Australian species
- Tradeoffs: Energetic costs of overwintering versus migrating vary across space



(eBird)

Flyways

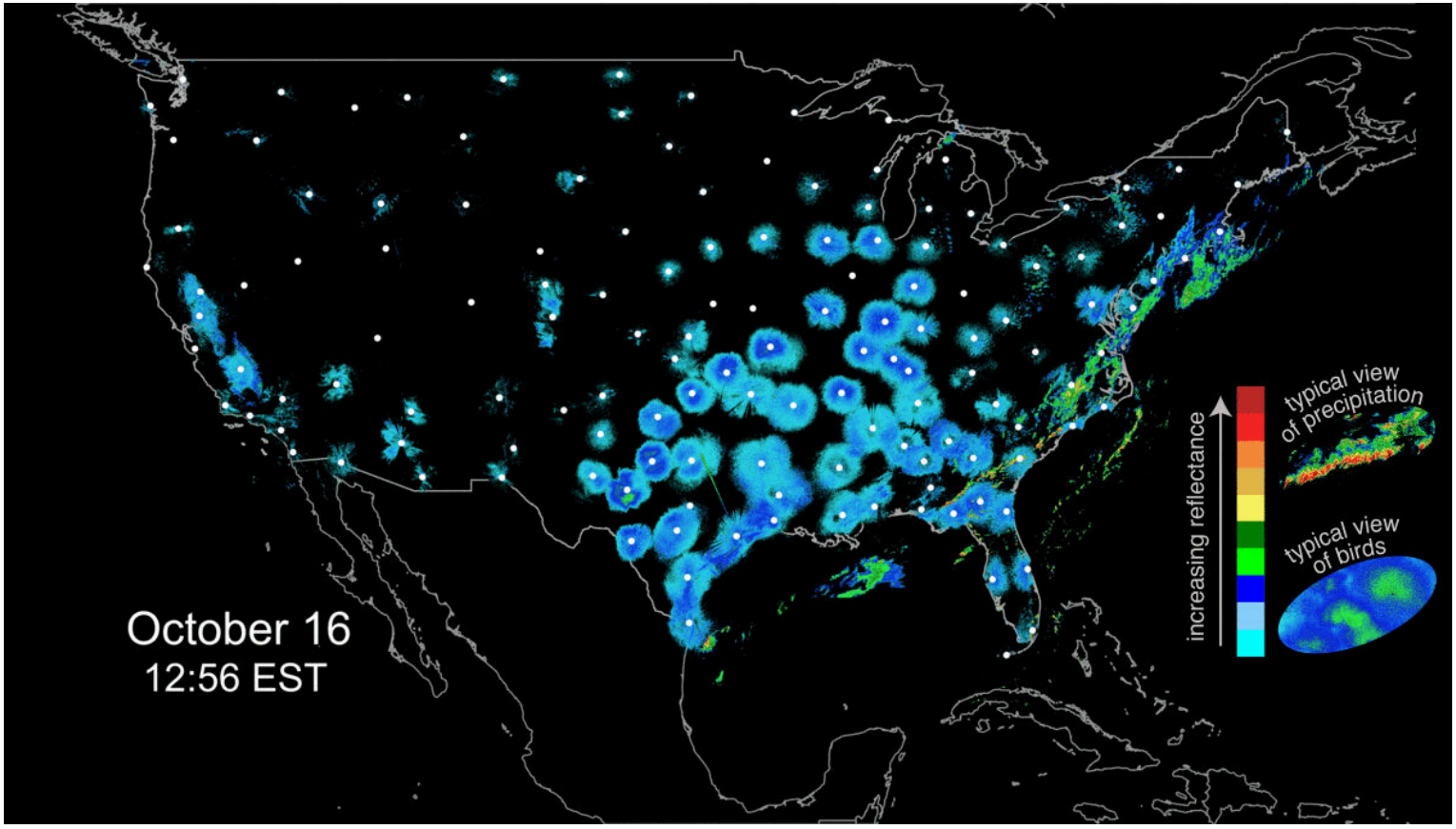
- **Flyways** are corridors where many species migrate
- can include **stopover sites** (think freezeout lake)



(eBird)

Radar ornithology

- radar can detect birds as well as weather events
- confused WWII era operators, who called them “angels”
- filtering out weather and other signals leaves us with migration forecasts!

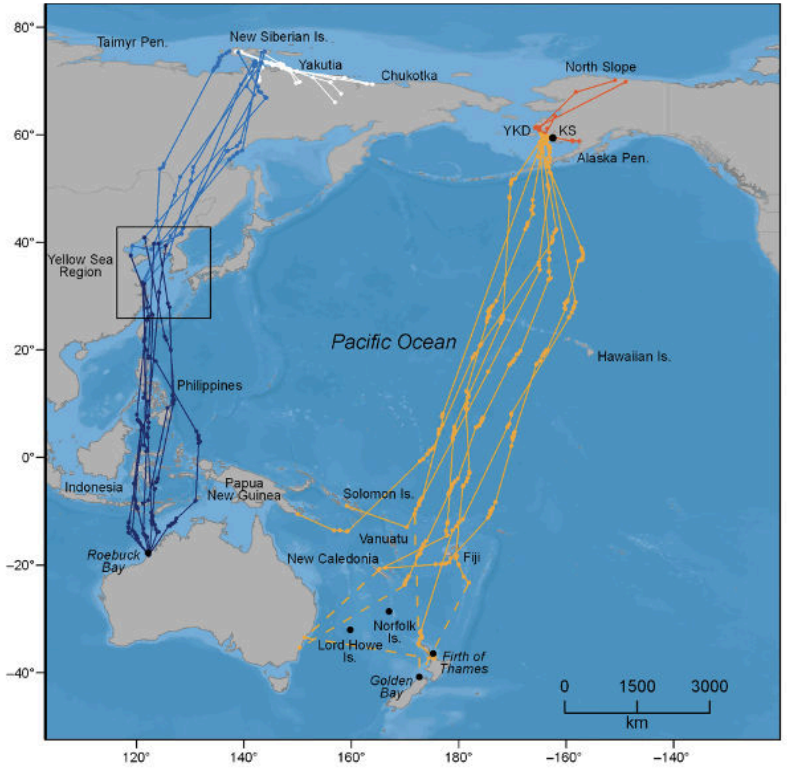


(Birdcast)

Direct tracking technology

- **geolocators** track daylight to log precise location of birds
- **radio telemetry** tracks birds from ground-based towers / hand-held devices
- **satellite tags** use GPS satellites

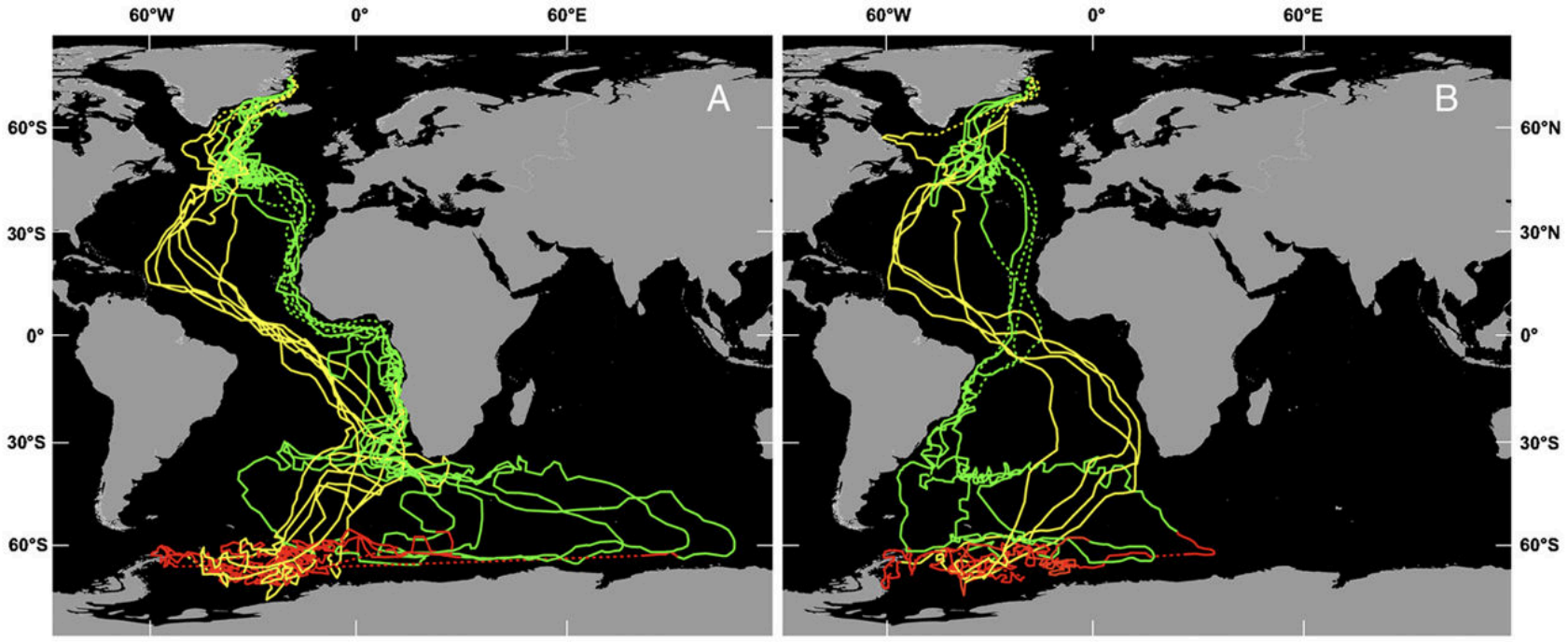




Conklin 2010; Battley et al. 2012

Migratory distances

- up to 96,000 km (59,000 mi) round trips for Arctic Terns
- up to 13,559 km (8425 mi) *nonstop* for Bar-tailed Godwits

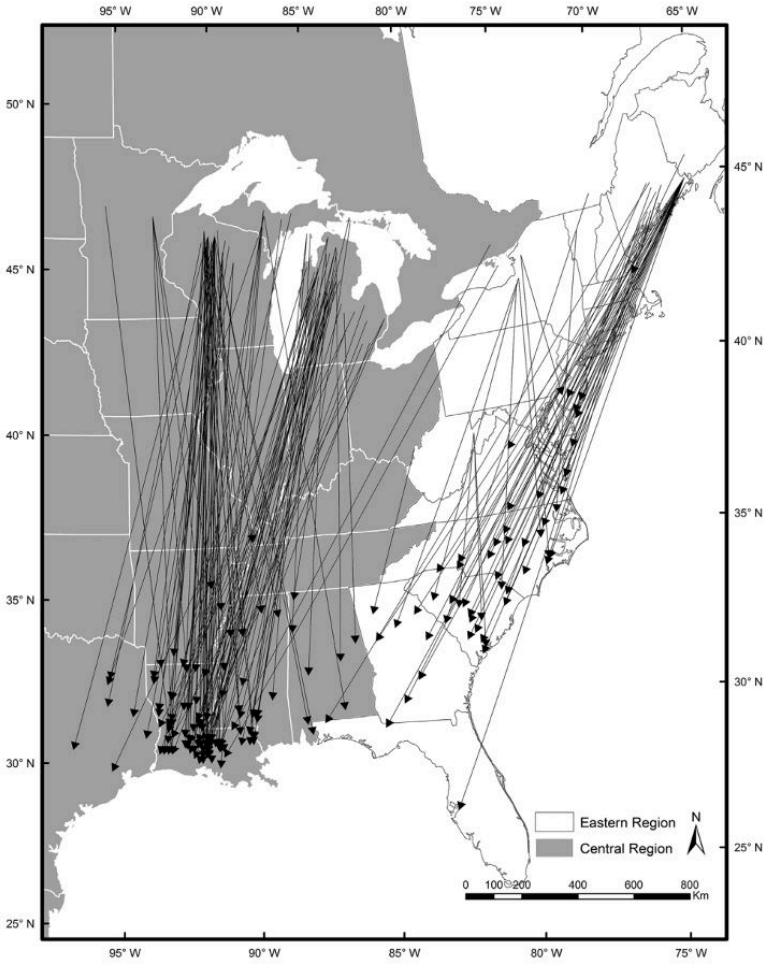


(Egevang et al. 2010)

Studying connectivity: bird banding

- Metal rings with unique codes attached to netted birds
- Long-term program by USGS (soon to be cut?)

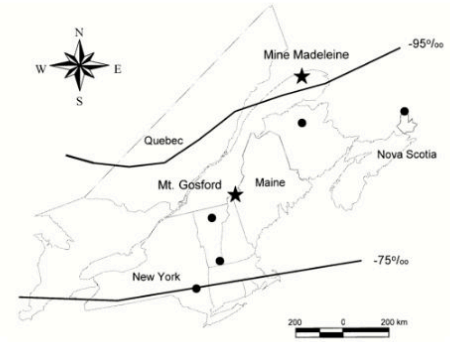




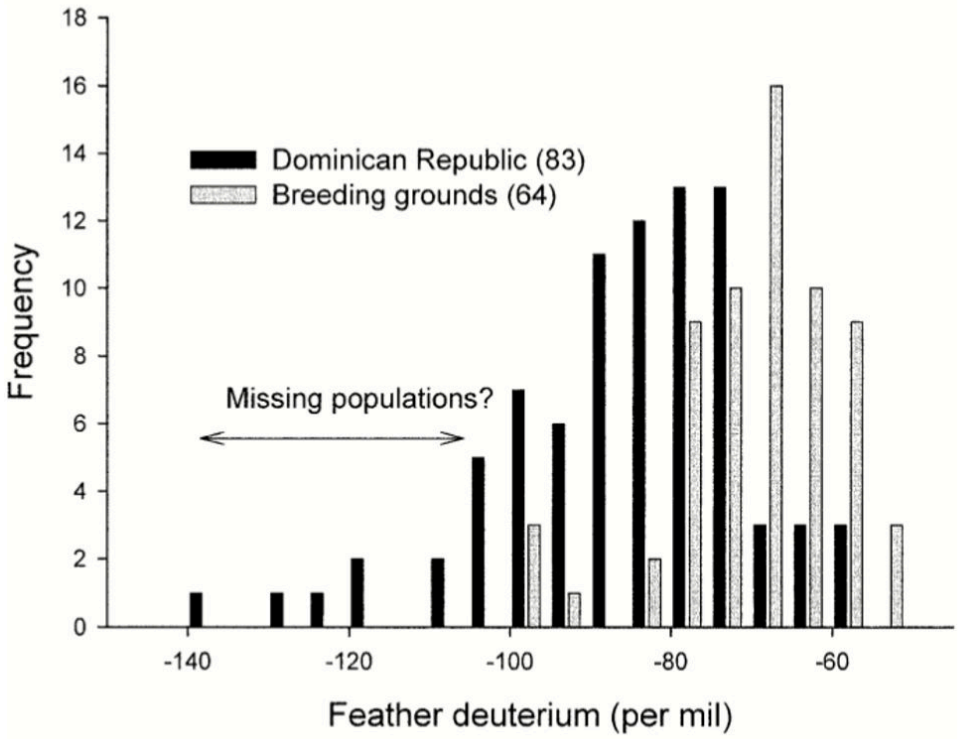
(USGS; Moore & Krementz 2017)

Stable isotopes

- deuterium (^2H) is a variant ("isotope") of hydrogen with same number of protons but different number of neutrons
- ratio of deuterium to hydrogen varies in precipitation—and thus feathers—across latitude and altitude
- can link breeding and wintering ranges



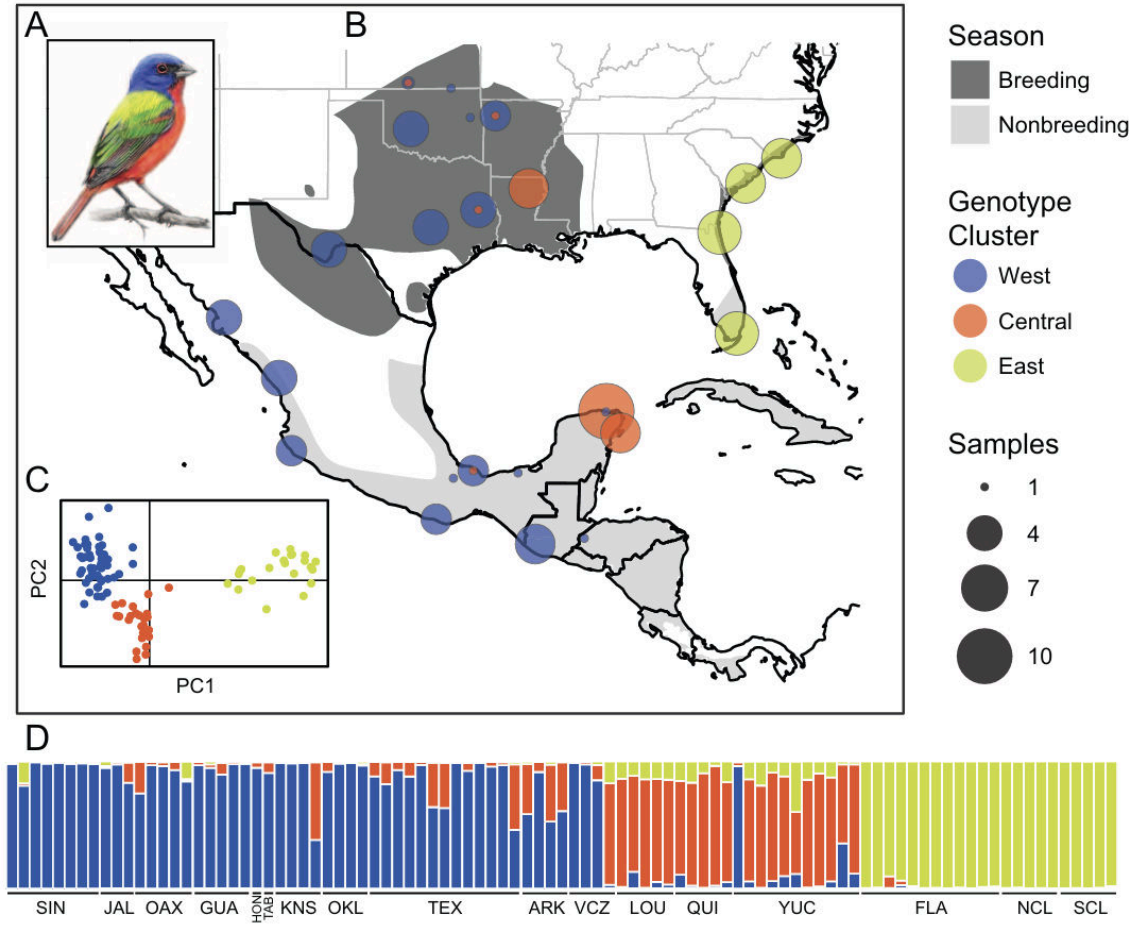




(*Catharus bicknelli*; Hobson et al. 2004)

Population genetics

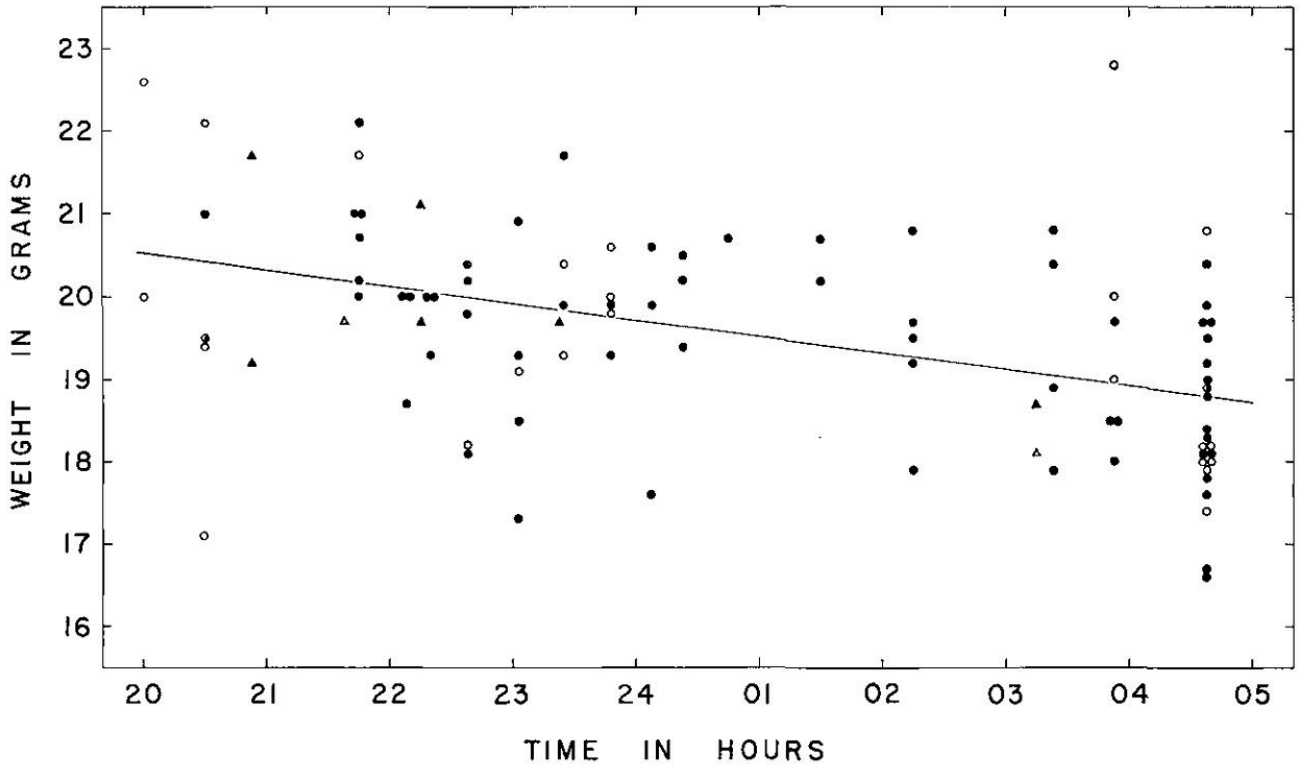
- connectivity can also be detected by similar DNA sequences / allele frequencies in wintering and breeding populations



(Battey et al. 2018)

Migration is costly

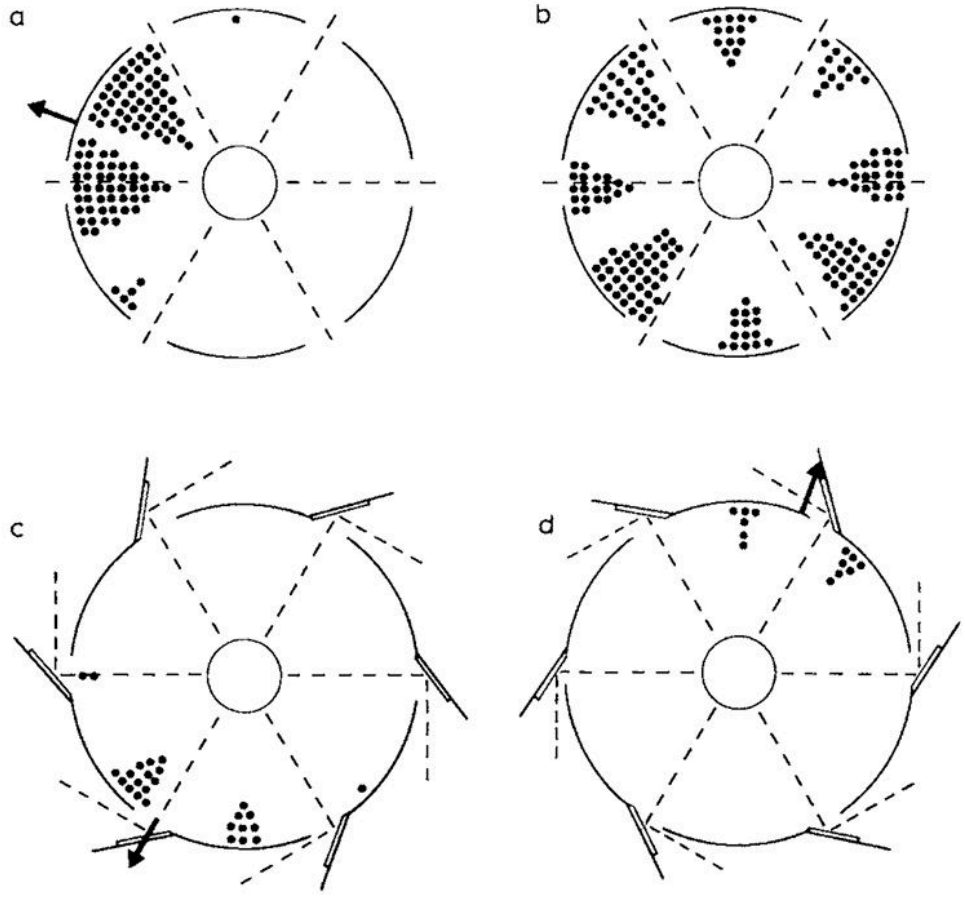
- Birds bird fat, carbohydrates, protein
- Ovenbirds lose 0.2g body weight / hour while migrating (16-20g total weight)



(Hussell 1969)

Navigation cues: sun

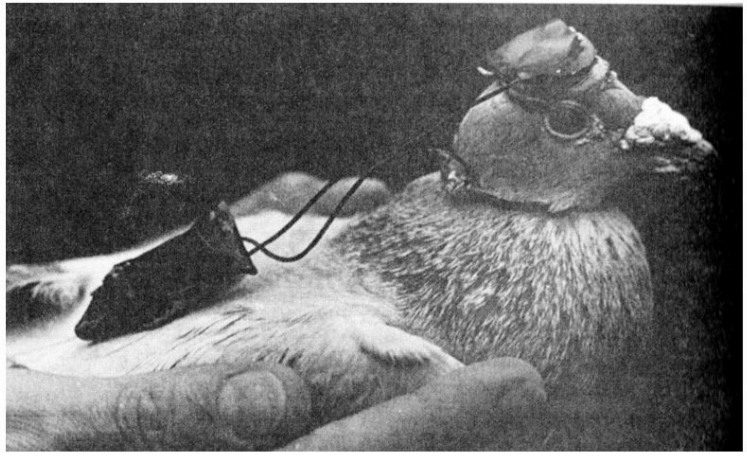
- Starlings change orientation during zugunruhe if sunlight is manipulated experiementally

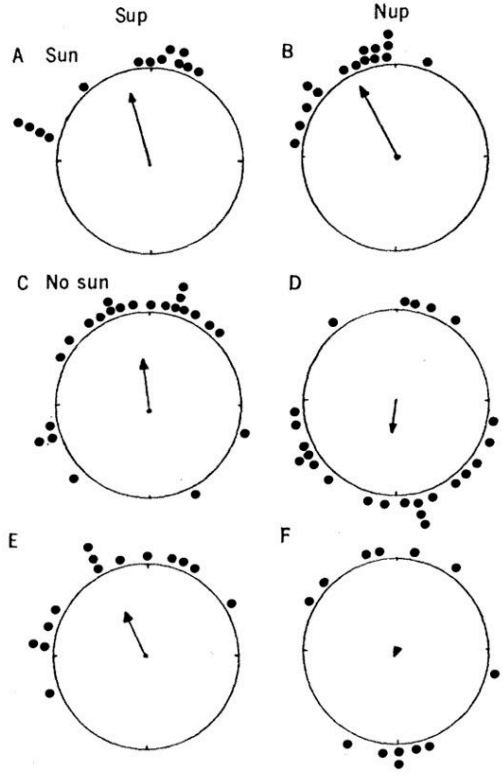


(Emlen 1975)

Migration cues: magnetic field

- Reversing the direction of an artificial magnetic field around a pigeon's head causes them to reverse their orientation on overcast days

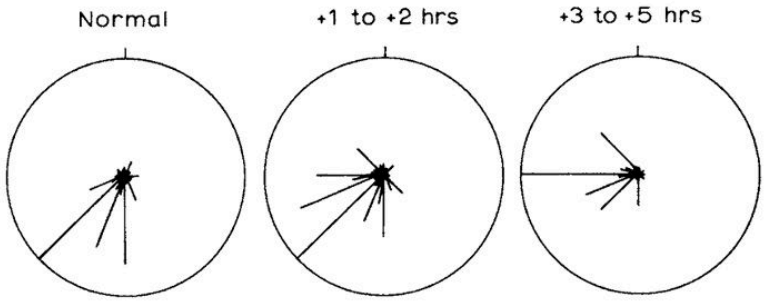




(Walcott & Green 1974)

Migration cues: stars

- Advancing the position of planetarium stars leads to orientation changes in nocturnal migrants





(*Sylvia atricapilla*; Emlen 1975)

Lecture Exam Review

- Wednesday, October 15th
- 50 points on feathers and flight, physiology, senses, vocalizations, annual cycles, and migration
- No lab content that wasn't also covered in lecture
- Question types: multiple choice, multiple select, true / false, fill-in-the-blanks, short answer
- Format: Canvas quiz, closed book. Bring a laptop or tablet.
- **Anything covered in lecture is fair game**

Feathers

- Identify feather types, tracts / apteria, major structures
- Name feather macromolecule
- Order feather developmental stages
- Describe the major hypothesis for feather evolution
- Match colors with pigment type or structural color / iridescence
- Describe fitness consequences of molt
- Contrast resident versus migrant molt patterns

Flight

- Explain how birds generate Bernoulli and Newtonian lift
- Define different types of drag
- Explain relationship between flight power, airspeed, and energetic cost
- Contrast thermal and dynamic soaring
- Identify major flight muscles and bones
- Predict ecology from wing aspect ration and wing loading

Physiology

- Relate bird body temperature to other vertebrates
- Describe respiratory system adaptations for flight
- Contrast basal metabolic rate, active metabolic rate, and metabolic scope
- Define thermoneutral zone
- Predict impact of summer plumage on BMR in winter
- Contrast nocturnal hypothermia, torpor and hibernation
- Describe adaptations to cope with hyperthermia
- Contrast mammalian and avian treatment of nitrogenous waste
- Explain importance of nasal salt glands

Senses

- Describe avian eye anatomy
- Contrast avian and human vision
- Explain cause and consequences of tetrahedral color space
- Explain how eye position, field of vision, and ecology relate
- Contrast avian and mammal ear anatomy
- Name organs responsible for tactile sensation
- Contrast avian and mammalian taste
- Describe basis and origin for carbohydrate taste in hummingbirds
- Link avian brain anatomy with intelligence
- Describe evidence for avian intelligence

Vocalizations

- Contrast avian larynx and syrinx
- Describe link between syrinx anatomy and vocal ability
- Explain the aspects of songs captured by amplitude and frequency
- Interpret spectrograms (e.g., which labeled section is louder / higher?)
- Name anatomical features responsible for harmonics, self-duetting, post-source modulation
- Explain impact of ecology on post-source modulation
- Evaluate evidence for acoustic adaptation hypothesis
- Name taxa that are vocal learners
- Recall phases of song learning and apply to mimicry / learning from “tutors”

Annual Cycles

- Contrast resident and migratory life cycles
- Explain link between food availability, molt, and migration
- Name environmental cues that mediate annnual cycles
- Define circadian and circannual rhythms
- Define zeitgebers and explain why they are necessary
- Explain the phenomenon of phenological mismatches

Migration

- Contrast hypotheses for evolution of seasonal migration
- Explain factors leading to partial migration
- Define flyways and stopover sites
- Evaluate pros and cons of different migration tracking methods
- Explain genetic and stable isotope evidence for migratory connectivity
- Describe energy sources for migration
- Describe evidence for nagivation by the sun, magnetic fields, and stars

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- Title slide species: *Acrocephalus taiti*, *Heteroxolmis dominicana*, *Momotus bahamensis*, *Poecile montanus*
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